FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FIITJEE

JEE(Main)-2015

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CRT –I
(Main)

S. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**A**	**A**	**C**
2.	**D**	**A**	**D**
3.	**B**	**D**	**B**
4.	**A**	**B**	**A**
5.	**C**	**B**	**D**
6.	**B**	**C**	**C**
7.	**B**	**B**	**B**
8.	**D**	**B**	**C**
9.	**A**	**A**	**A**
10.	**C**	**C**	**C**
11.	**B**	**B**	**C**
12.	**D**	**D**	**D**
13.	**C**	**C**	**B**
14.	**B**	**D**	**B**
15.	**D**	**C**	**A**
16.	**C**	**A**	**B**
17.	**B**	**D**	**B**
18.	**A**	**B**	**B**
19.	**B**	**B**	**C**
20.	A	**A**	**B**
21.	D	**A**	**C**
22.	**D**	**C**	**B**
23.	**C**	**D**	**C**
24.	**A**	**B**	**B**
25.	**A**	**C**	**C**
26.	**C**	**C**	**D**
27.	**B**	**B**	**B**
28.	**D**	**B**	**D**
29.	**A**	**B**	**B**
30.	**A**	**D**	**C**

Physics

PART – I

SECTION – A

1. No. of stations = $\frac{\text{Total Bandwidth}}{\text{BW per statation}} = \frac{100\times10^3}{10\times10^3} = 10.$

3. B,



Applying conservation of momentum : $2mv = mv_0$ . .. (i)

conservation of energy : $2\times\frac{1}{2}mv^2 + \frac{1}{4\pi\varepsilon_0}\frac{q^2}{d} = \frac{1}{2}mv_0^2$ . . . (ii)

(I) $\Rightarrow$ $v = v_0/2$, putting in (2), we get,

$d = \frac{q^2}{\pi\varepsilon_0 mv_0^2}$

4. the powers of the lenses add, and the power of each half is one half of that of the entire lens.

5. $\beta = \frac{\partial I_C}{\partial_B} = 100$, $R_i = \frac{\partial V_B}{\partial I_B}$

power gains = $\beta^2\frac{R_L}{R_i} = 5\times10^4$

6. $I_A = \frac{I_0}{2}$

$I_C = \frac{I_0}{2}\cos30° = \frac{I_0}{2}\frac{3}{4} = \frac{3I_0}{8}$

$I_B = I_C\cos^2 60° = \frac{3I_0}{8}\cos^2 60° = \frac{3}{32}I_0$

$\frac{I_B}{I_C} = \frac{3}{32} = 9.4\%$

7. $i = \frac{5}{5+R}$

$B = B_H\tan\theta$

$\frac{\mu_0 Ni}{2r} = B_H\tan\theta$

$i = \frac{2rB_H}{\mu_0 N}\tan\theta = \frac{5}{5+R}$

Solving $R = 3.5\,\Omega$

9. Since the current flowing through a loop is determined by the net emf acting in the loop, which again is zero n this case - as the flux doesn't change.

10. At x = 31.5 mm, both 3600 A^0 and 6000 A^0 produce minima. If the slit separation is d, then,

$$\frac{xd}{D\lambda_1} = n_1 + \frac{1}{2}, \qquad \frac{xd}{D\lambda_2} = n_2 + \frac{1}{2}$$

or , $\frac{xd}{D}\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right) = n_1 - n_2 = 7$

Substituting the values, D = 1m, λ_1 = 3600 A^0, λ_2 = 6000 A^0
we get d = 0.2 mm.

11. $\frac{dN_A}{dt} = (-\lambda N_A) + (-2\lambda . N_A) + (-3\lambda . N_A)$

$= -6\,\lambda .\ N_A$

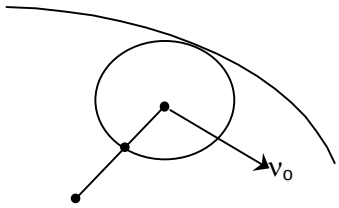
12. The mass of the liquid does not change, the area on which it exerts its weight increases by $2 \times 10^{-3} \times 10$ or 2×10^{-2}
or 2 %, the pressure decreases by 2 %.

13. At the farthest point X,
$mv_0R = mv_x\,4R$

$$\frac{1}{2} m v_x^2 - \frac{GMm}{4R} = \frac{1}{2} m v_0^2 - \frac{GMm}{R}$$

$\Rightarrow$ $v_0 = \sqrt{1.6gR}$.

14. The angular velocity ω of the sphere about an axis through its C.M. is $\frac{v_0}{d/2} = \frac{2v_0}{d}$, it does not depend on whether the axis is accelerated or not.



15. Their average translational K.E. is $\frac{3}{2} k_B T$, $k_B \rightarrow$ Boltzmann's constant

$$\frac{1}{2} m_{H_2} C_{H_2}^2 = \frac{1}{2} m_{He} C_{He}^2 = \frac{3}{2} k_B T$$

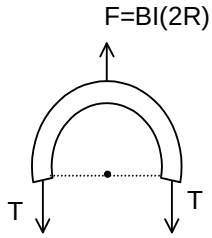
$C_{H_2} : C_{He} = \sqrt{m_{He}} : \sqrt{m_{H_2}} = \sqrt{4} : \sqrt{2} = 2 : \sqrt{2}$

16. Using $E_{k\alpha} = \frac{3}{4} 13.6 \text{ eV } (Z-1)^2$ & $\lambda_{k\alpha} = \frac{12400}{E_{k\alpha}} A^0$

17. For the mixture, $\gamma = \frac{C_P^{(1)} + C_P^{(2)}}{C_V^{(1)} + C_V^{(2)}} = \frac{\frac{7}{2}R + \frac{5}{2}R}{\frac{5}{2}R + \frac{3}{2}R} = \frac{3}{2}$

Applying $TV^{\gamma - 1}$ = constant, the temperature increases by a factor of $\sqrt{8}$.

20. (A), Consider a semicircular portion of the wire



$2T = BI.\ 2R$

$\Rightarrow \quad T = BIR$

21. Time to reach B from A = $t_{AB} = \int_{x=0}^{d} \frac{dx}{c(x)}$

$= \frac{d}{0.1c_0} \ln (1.1)$; total time = $2t_{AB}$

22. For the system,

$\frac{mv^2}{r} = Bev$

$mvr = \frac{nh}{2\pi}$

$\Rightarrow \quad r = \sqrt{\frac{nh}{Be}}$.

23. $pV^{1/4}$ = constant

$\Rightarrow \quad \frac{RT}{V} v^{1/4}$ = constant

or $TV^{-3/4}$ = constant

$dT\ V^{-3/4} - \frac{3}{4} dV.V^{-7/4}.T = 0$

or $\frac{dV}{dt} = \frac{4}{3}\frac{V}{T}$

$C = \frac{dQ}{dT} = \frac{dU}{dT} + P\frac{dV}{dT} = C_v + p.\frac{4}{3}\frac{V}{T} = \frac{5}{2}R + \frac{4}{3}R = \frac{23}{6}R$.

25. Time constant = RC, increases with temperature

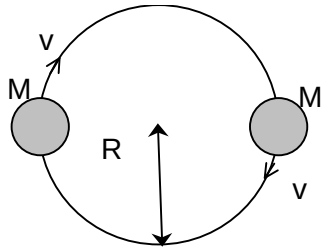
27. $\frac{dN_B}{dt} = \lambda_A N_A - \lambda_B N_B = 4\lambda N_A - 9\lambda N_B = 0$ (N_B is maximum)

$\therefore \quad \frac{N_A}{N_B} = \frac{9}{4}$

29. For the position -electron system, the mass of the position = mass of the electron, much less than that of a proton . the distance in the ground state > a_B (hydrogen atom) A careful calculation shows $a \neq 1.06\ A^0$ (= 2 a_B)

30. (A), $M\frac{v^2}{R} = \frac{GM^2}{4R^2}$

$$v^2 = \frac{GM}{4R}$$



Time period, $T = \frac{2\pi R}{v} = \frac{2\pi R}{\sqrt{GM/4R}}$

$= \frac{2\pi}{\sqrt{GM}} \times 2 \times R^{3/2}$

If $M = M_{sun}$, $R = R_{earth\text{ - }sun,}$ Kepler's 3rd law gives us,

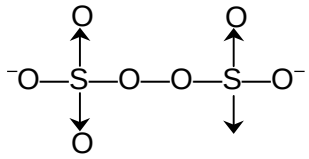
$T_{earth} = \frac{2\pi}{\sqrt{GM_{sun}}} \times R^{3/2}_{earth-sun} = 1$ yr.

$\therefore \quad T = \frac{2\pi}{\sqrt{GM_{sun}}} \times 2 \times R^{3/2}_{earth-sun} = 2 \times 1$ yr $= 2$yr .

Chemistry

PART – II

SECTION – A

2. 

4. i for $AgNO_3$ = $\frac{\text{normal molecular weight}}{\text{observed molecular weight}} = 1 + \alpha$

$\alpha = \frac{170}{92.64} - 1 = 0.835 = 83.5\%$

5. At 25°C $[H^+] = 10^{-7}$; $\therefore K_w = 10^{-14}$
At 35°C $[H^+] = 10^{-6}$; $\therefore K_w = 10^{-12}$
Now using,

$2.303 \log \frac{K_{W_2}}{K_{W_1}} = \frac{\Delta H}{R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$

We get, $\Delta H = 84.55$ kcal/mole

Thus $H_2O \rightleftharpoons H^+ + OH^-$ $\Delta H = 84.55$

$\therefore H^+ + OH^- \rightleftharpoons H_2O$ $\Delta H = -84.55$ kcal/mole

6. No. of atoms of A from corners of unit cell = $\frac{7}{8}$

No. of atoms of B from faces of unit cell = 3

Thus A : B : $\frac{7}{8}$: 3 (or) 7 : 24

$\therefore$ (C)

12. $^-O-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-CF_3$ is the best leaving group as it is the weakest base

14. Because it has maximum no. of unpaired electrons

16. Weight of H_2O_2 in 1 ml = $\frac{17}{1120}$

Moles of H_2O_2 in 1 ml = $\frac{17}{1120} \times \frac{1}{34}$

Moles of O_2 liberated = $\frac{17}{1120} \times \frac{1}{34} \times \frac{1}{2}$

Volume of O_2 liberated = $\frac{17}{1120} \times \frac{1}{34} \times \frac{1}{2} \times 22400$ = 5 ml

17. $PM = dRT$ or $d \propto \frac{1}{T}$

$$\frac{d_1}{d_2} = \frac{T_2}{T_1} = \frac{d}{0.75d} = \frac{T_2 + 273}{27 + 273}$$

$\therefore T_2 = 127°C$

18. $$\frac{0.693}{t_{1/2}} = \frac{2.303}{t} \log \frac{A_0}{A_t}$$

Putting the values we get

$t_{1/2} = 0.693$

19. $pH = 12$, $pOH = 2 \Rightarrow [OH^-] = 10^{-2}$ M

20. % of M in $M_xO_y = \frac{xM \times 100}{xM + 16y} = 60 \therefore xM = 24y$

$$\% \text{ M in } M_xBr_{2y} = \frac{xM}{xM + 160y} \times 100 = \frac{24y}{24y + 160y} \times 100 \approx 13$$

$\therefore$ % of bromine in $M_xBr_{2y} \cong 87$

21. Equivalents of $CaCO_3 = \frac{50 \times 2}{100} = 1$

Equivalent of HCl $= \frac{60 \times 20}{1000} = 1.2$

$\therefore$ Equivalent of $CO_2 = 1$, moles of $CO_2 = 0.5$ (n-factor = 2)

$\therefore$ mass of $CO_2 = 22g$

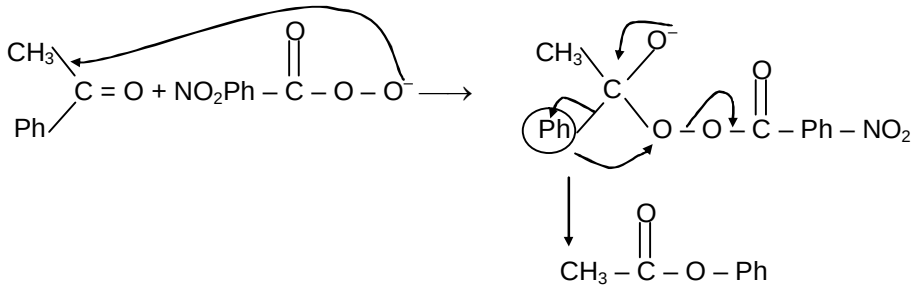
22. $$HA + OH^- \underset{Kh}{\overset{Ka}{\rightleftharpoons}} A^- + H_2O$$

$$K_h = \frac{K_w}{K_a} = \frac{10^{-14}}{10^{-5}} = 10^{-9}$$

$K_a = K_h^{-1} = 10^9$

23. Ag has a lower ionisation potential than Cu because of larger size. But in Au due poor shielding effect of f-electrons the nuclear charge is not shielded and so I.P. is high

$\therefore$ Au > Cu > Ag

24. 

27. m moles of KI = 50 × 0.04 = 2
m.eq. of $Na_2S_2O_3$ reacted with 10 ml solution = $20 \times 2.5 \times 10^{-3}$
$= 50 \times 10^{-3}$
m.eq. of I^- left unreacted in 100 ml solution = $50 \times 10^{-3} \times \frac{100}{10} = 0.5$
m. moles of I^- reacted with $CuCO_3$ = 2 – 0.5 = 1.5
$\because 2Cu^{++} + 5I^- \longrightarrow 2CuI + I_3^-$
m.moles of Cu^{++} = $\frac{2}{5} \times 1.5 = 0.6$
wt. of $CuCO_3$ = $0.6 \times 10^{-3} \times 123.5$
= 0.0741
wt. of $CuCO_3$ = $\frac{0.0741}{0.1} \times 100 = 74.1$

28. $2NH_2 + CO_2 \longrightarrow NH_2 - CO - NH_2 + H_2O$
moles of urea = $\frac{999.6 \times 10^3}{60} = 16.66 \times 10^3$
$\therefore$ moles of $CO_2 = 16.66 \times 10^3$
$C_nH_{2n+2} + O_2 \longrightarrow nCO_2 + H_2O$
$\frac{16.66 \times 10^3}{n} \times \left[12n + (2n+2)\right] = 236 \times 10^3$
$\Rightarrow n \cong 12$
so, M. F. of hydrocarbon is $C_{12}H_{26}$

29. $K_{eq}(300\ K)\ \frac{K_f}{K_b} = \frac{A.e^{-(E_a)_f/RT}}{A.e^{-(E_a)_b/RT}} = \frac{0.8}{0.2} = 4$
$\therefore \ln \frac{k_2}{k_1} = \frac{\Delta H}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$
$\ln\left(\frac{4}{k_2}\right) = 1$
$\frac{4}{k_2} = e \Rightarrow k_2 = 4e^{-1}$

30. In hydrazine ($NH_2 - NH_2$) bond order is one, whereas in nitrogen gas bond order is three.

Mathematics

PART – III

SECTION – A

1. $f(x) = e^x (\ln x)^{2n}$

$\lim\limits_{n\to\infty} f(x) = 0$, if $x \in \left(\frac{1}{e}, e\right)$

$\therefore \int\limits_{1/e}^{e} g(x)dx = 0$

2. Clearly $f(x) = x^4$

$\therefore$ Required area $= 2\int\limits_0^1 \left(\sqrt{2-x^2} - x^4\right)dx = \frac{\pi}{2} + \frac{3}{5}$

3. $I_1 = \int\limits_{\sin^2 t}^{1+\cos^2 t} x f\{x(2-x)\}dx = 2I_2 - I_1$

$\Rightarrow \frac{I_1}{I_2} = 1$

4. $f'(x) = \frac{f(x)}{6f(x) - x}$

$\therefore I = -\int \frac{2x - f'(x)}{\left(x^2 - f(x)\right)^2} dx = \frac{1}{x^2 - f(x)} + c$

5. $2x + \frac{y}{8} = \alpha$, $2x - \frac{y}{8} = \beta$

$\therefore f(\alpha, \beta) = \alpha^2 - \beta^2$

$\therefore f(m, n) + f(n, m) = 0$

6. $\frac{\pi}{\sqrt{k}} = \alpha \Rightarrow k = 1 \Rightarrow \lim\limits_{n\to\infty} k^n = 1$

7. $g'(x) = x(f'(2x^2 - 1) - xf'(1 - x^2))$

If, $x > 0 \Rightarrow x \in \left(\sqrt{\frac{2}{3}}, \infty\right)$

If, $x < 0 \Rightarrow x \in \left(-\sqrt{\frac{2}{3}}, 0\right)$

$\therefore x \in \left(-\sqrt{\frac{2}{3}}, 0\right) \cup \left(\frac{\sqrt{2}}{3}, \infty\right)$

8. $\sin\theta - \cos\theta = \frac{b}{a}$

$\sin\theta\cos\theta = \frac{c}{a}$

$\therefore\ b^2 = a^2 - 2ac$

$\therefore\ \cos B = \frac{a^2 + c^2 - b^2}{2ac} = 1 + \frac{c}{2a}$

9. $\frac{x^3}{3} - 4x^2 + 13x = x\sin\left(\frac{a}{x}\right)$

$\Rightarrow \frac{1}{3}\left\{(x-6)^2 + 3\right\} = \sin\frac{a}{x}$

$\therefore\ \sin\frac{a}{6} = 1$

10. $\because\ \vec{x}, \vec{y}$ and $\vec{x}\times\vec{y}$ are linearly independent

$\therefore\ 20a - 15b = 15b - 12c = 12c - 20a = 0$

$\therefore\ c^2 = a^2 + b^2$

11. $\vec{r} = \left(\hat{i} + 2\hat{j} + 3\hat{k}\right) + \lambda\left(\hat{i} - \hat{j} - 2\hat{k}\right) + \mu\left(-\hat{i} + 2\hat{k}\right)$

$\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$

$\vec{b} = \hat{i} - \hat{j} - 2\hat{k}$

$\vec{c} = -\hat{i} + 2\hat{k}$

$\therefore$ Vector equation of plane is $(\vec{r} - \vec{a})(\vec{b}\times\vec{c}) = 0$

$\Rightarrow 2x + z = 5$

12. $\therefore\ a > 1$

$\therefore\ b^2 - 5b + 7 > 1$

$\therefore\ b \in (-\infty, 2) \cup (3, \infty)$

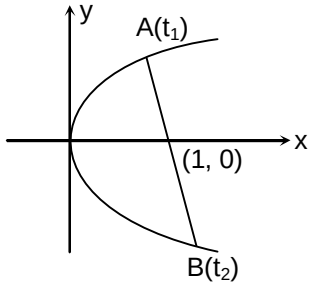
13. Clearly, $-\lambda < 0$, $\lambda > 0$

14. $m_{AB} = \frac{2}{t_1 + t_2} = 1$

$t_1 + t_2 = 2$

$t_1t_2 = -1$

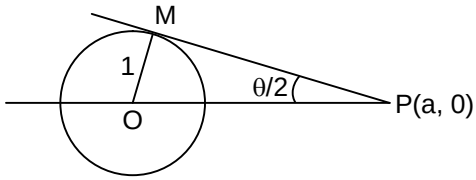
$\therefore$ Required equation is $m^2 + 2m - 1 = 0$



15. $\frac{\pi}{3} < \theta < \pi$

$\Rightarrow \frac{\pi}{6} < \frac{\theta}{2} < \frac{\pi}{2} \Rightarrow \frac{1}{2} < \frac{1}{a} < 1 \Rightarrow 1 < a < 2$

$\therefore\ a \in (-2, -1) \cup (1, 2)$



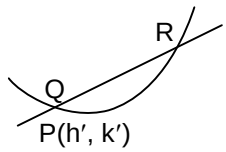
16. $x = m(y-10) - \frac{1}{m}$

$x - h = m(y-k) + \frac{a}{m}$

$\therefore$ h = 0, k = 10, a = –1

$\therefore (x-0)^2 = -4(y-10)$

17. $\therefore \frac{x-h'}{\cos\theta} = \frac{y-k'}{\sin\theta} = r$



$\therefore PQ \cdot PR = \frac{ah'^2 + 2hh'k' + bk'^2}{a\cos^2\theta + 2h\sin\theta\cos\theta + b\sin^2\theta}$

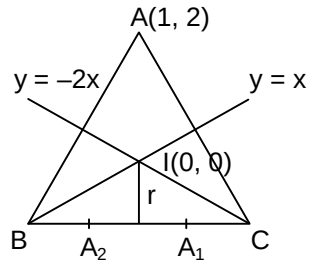
$\therefore$ a = b, h = 0

18. $A_1 = (2, 1)$

$A_2 = \left(-\frac{11}{5}, \frac{2}{5}\right)$

Equation of BC: x – 7y + 5 = 0

$\therefore r = ID = \left|\frac{5}{\sqrt{1+49}}\right| = \frac{1}{\sqrt{2}}$



19. $R_1 \to R_1 - R_2, R_2 \to R_2 - R_3$

$\Delta = [x] + [y] + [z] + 1$

$\therefore \Delta_{max} = 1 + 0 + 2 + 1 = 4$

20. $-\left\{\frac{(a-b)^2 - 4(1-a-b)}{4\times 1}\right\} > -\left\{\frac{(a+b)^2 - 4(1+a+b)}{4\times(-1)}\right\}$

$\Rightarrow a^2 + b^2 < 4$

21. Put, $\cot^{-1} x = \theta$

$\therefore e^{2i\cot^{-1}x} = \cos 2\theta + i\sin 2\theta = \frac{xi-1}{xi+1}$

$\Rightarrow e^{2mi\cot^{-1}x}\left(\frac{xi+1}{xi-1}\right)^m = 1$

22. $\frac{5^{200}}{8} = \frac{(1+24)^{100}}{8}$

$\therefore \left\{\frac{5^{200}}{8}\right\} = \frac{1}{8}$

23. $\frac{1}{a_4} - \frac{1}{a_3} = \frac{1}{a_2} - \frac{1}{a_1}$ ..... (1)

$3\left(\frac{1}{a_3} - \frac{1}{a_2}\right) = \frac{1}{a_4} - \frac{1}{a_1}$ ..... (2)

$\therefore$ From (1) and (2), we get $\frac{1}{a_2} - \frac{1}{a_1} = \frac{1}{a_3} - \frac{1}{a_2} = \frac{1}{a_4} - \frac{1}{a_3}$

$\therefore a_1, a_2, a_3, a_4$ are in H.P.

24. ${}^{n}C_2 = \frac{n(n-1)}{2}$ = Number of possible pairs of n objects

Total weight of $\frac{n(n-1)}{2}$ pairs are $\frac{n(n-1)}{2} \times 2 \times w = n(n-1)w$ units

$\therefore$ n(n – 1)w = 120 ..... (1)

Similarly, total weight of all triplets = 480

$\therefore \frac{n(n-1)(n-2)3w}{6} = 480$ ..... (2)

$\therefore$ From (1) and (2), we get n = 10

25. $\left|\vec{a} + 2\vec{b} + 3\vec{c}\right|^2 = 3 + 2\sqrt{2}$

4 cos α + 6 cos β = –11 + $2\sqrt{2}$ – 12 cos γ (γ is angle between $\vec{b}$ and $\vec{c}$)

$\therefore$ 4 cos α + 6 cos $\beta \in \left[-11+2\sqrt{2}, -5+2\sqrt{2}\right]$

26. $P\left(\frac{B}{A \cup B^C}\right) = \frac{P(A \cap B)}{P\left(A \cup B^c\right)} = \frac{0.2}{0.8} = \frac{1}{4}$

27. Let $\alpha = |\underline{MAP} = \cot^{-1}(3.2)$

and $\beta = |\underline{MBP} = \text{cosec}^{-1}(2.6)$

$\Rightarrow \cot\alpha = 3.2$ and $\text{cosec}\beta = 2.6$

Let height of the tower = h

$\Rightarrow$ MP = h.

Now BM = MP $\cot\beta$ = h $\cot\beta$

= $h\sqrt{\text{cosec}^2\beta - 1} = h\sqrt{(2.6)^2 - 1} = (2.4)h$

and AM = MP $\cot\alpha$ = h $\cot\alpha$ = h(3.2) = 3.2 h

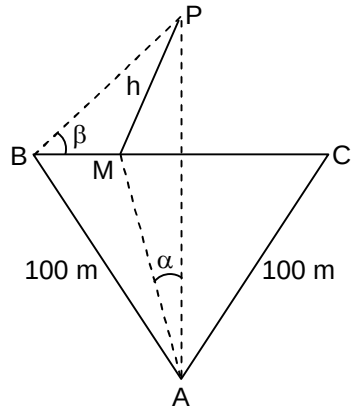
Hence, from ΔABM

$AB^2 = AM^2 + BM^2$

$(100)^2 = (3.2h)^2 + (2.4h)^2$

$h^2 = \frac{100^2}{10.24 + 5.76} = \frac{100^2}{16}$

$h = \frac{100}{4} = 25$.



28. We have,

bN = {bx | x $\in$ N } = the set of positive integral multiples of b

cN = {cx | x $\in$ N } = the set of positive integral multiple of c

$\therefore$ bN $\cap$ cN = the set of positive integral multiples of L.C.M of b and c

$\Rightarrow$ d = L.C.M. of b and c

29. $f'(0) = -\frac{1}{3}$

$\therefore$ Required limit = $\lim_{x \to 0} \frac{2x}{2xf'\left(x^2\right) - 40xf'\left(4x^2\right) + 56xf'\left(7x^2\right)} = -\frac{1}{3}$

30. P(1), P(2) and P (3) are not true

P (4) is true and $2^k < k!$

$\Rightarrow 2.2^k < 2k! \le (k+1)k!$ for $k \ge 1$